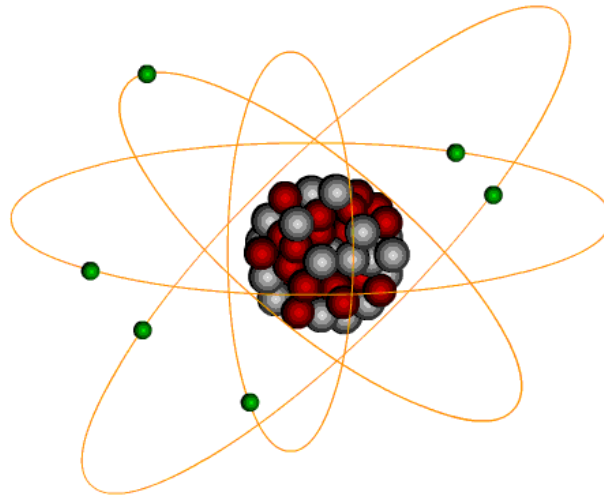


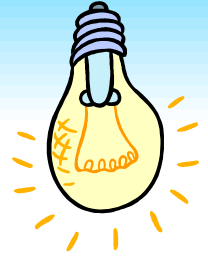


Black Body Radiation

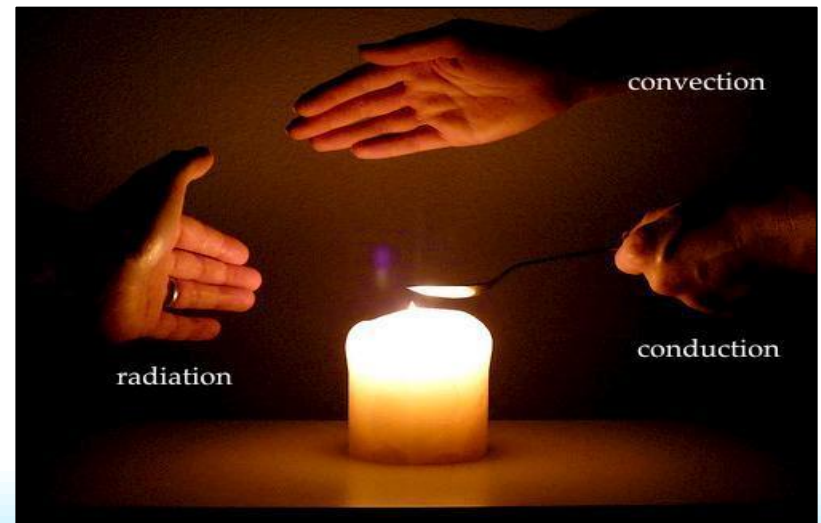
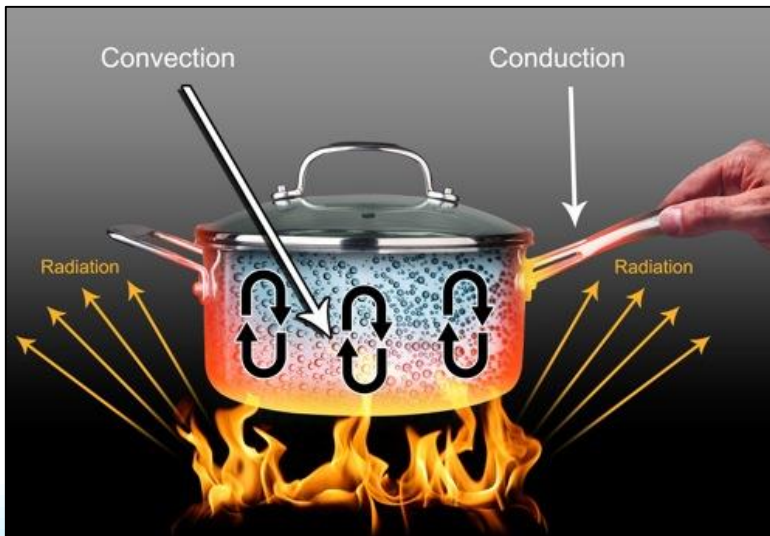


Sri Lanka Atomic Energy Board (SLAEB)

Radiation

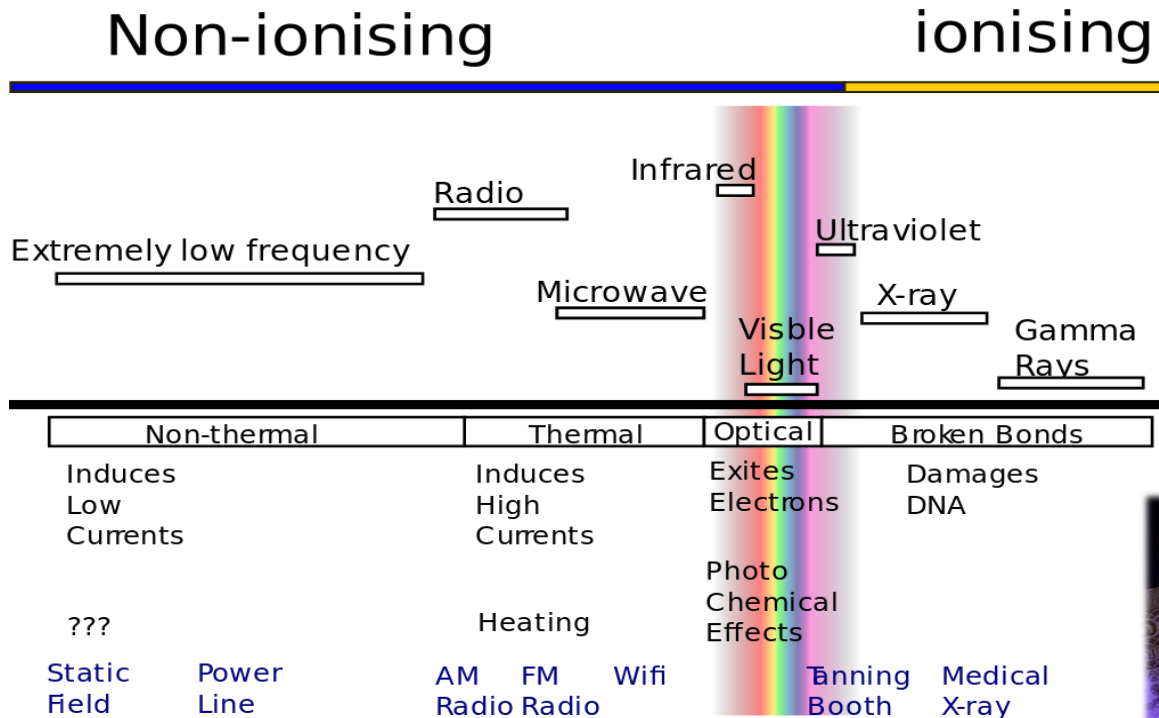
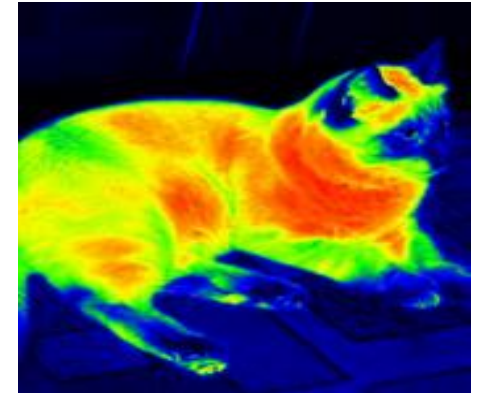


- Radiation is the transfer of thermal energy by electromagnetic (EM) waves.
- Conduction and convection require a medium to transfer thermal energy.
- Radiation does not require any medium.
- It can take place in a vacuum.



Radiation (cont.)

- Waves such as visible light, infrared, and ultraviolet light are examples of radiation.



EM Spectrum

Infrared/IR Waves

- A part of the EM spectrum.
- Wave length range is 750 nm to 1 mm.
- The main source of IR radiation is through temperature or thermal radiation.
- The more heat there, the more movement the atoms and molecules have, and the more infrared radiation is produced.

(The frequency of IR waves and the vibration frequency of molecules are same. Hence, due to resonance the temperature of object increase rapidly)

Infrared/IR Waves (cont.)

Environmental Effects of IR Radiation – Greenhouse Effect

- Objects, the surfaces, and bodies of water are heated by the visible light of the sun.
- They emit IR radiation.
- Gases like water vapor in the atmosphere absorb them.
- As a result of this heat the atmosphere.



Infrared/IR Waves (cont.)

Applications

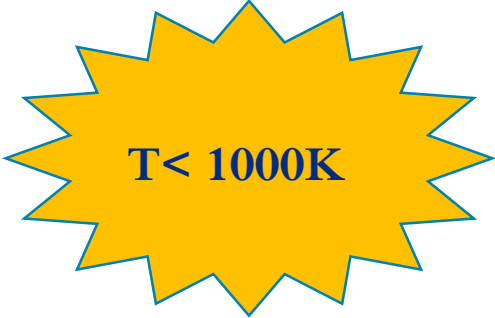


Thermal cameras and imaging systems respond to infra-red radiation and, suitably calibrated for emissivity, can produce images of the surface temperatures of bodies.

- Medical applications,
- Military applications, etc.

Thermal Radiation

- Thermal radiation is emitted by all objects above absolute zero (0 K), because of the thermal motion of atomic particles.
- Thermal radiation rate ($\uparrow$) depends on;
 - Surface temperature of the object $\uparrow$
 - Surface area of the object $\uparrow$
 - Type of the surface of the object



$T < 1000\text{K}$

IR

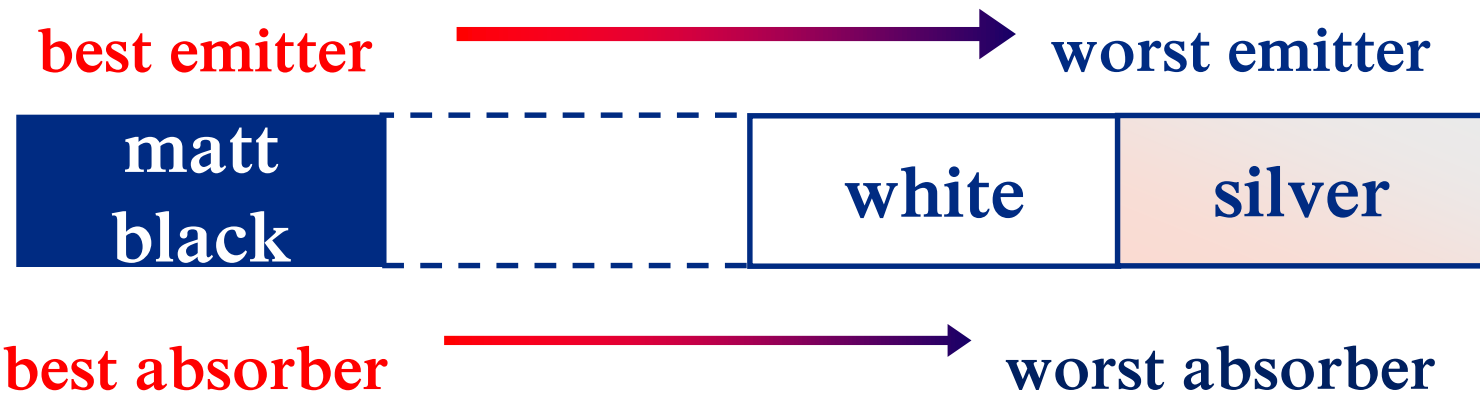


$T > 1000\text{K}$

IR, visible, UV

Thermal Radiation (cont.)

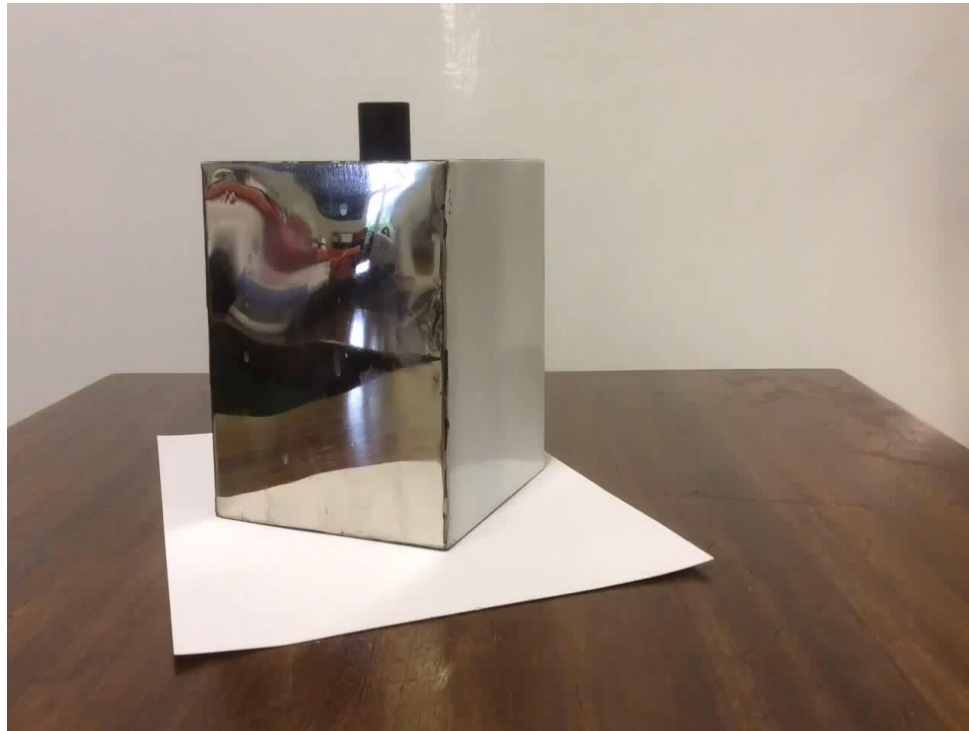
- All objects both absorb and emit radiation, though some are better than others.
- Good emitters are also good absorbers.



- Matt black surfaces are the best absorbers of radiation.
- Shiny surfaces are the worst emitters because they reflect most of the radiation away.



The Rate of Thermal Radiation Emission

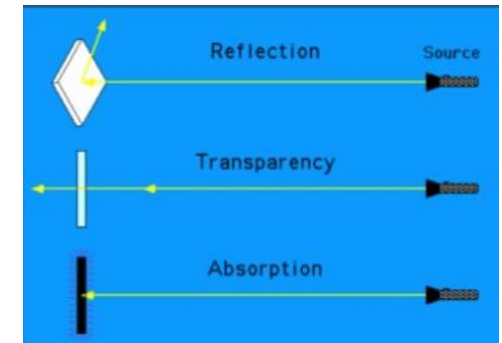


Diathermanous and Adiathermanous Mediums

Light Rays

- The medium will allow the light rays to pass through it → Transparent medium

- The medium will absorb all incident light rays → Opaque medium



Thermal Rays

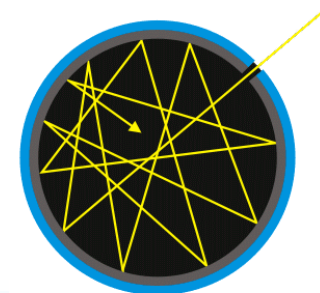
- The medium will allow the thermal rays to pass through it → Diathermanous

- The medium will absorb all incident thermal rays → Adiathermanous

Black Body (Ideal Thermal Radiator)

- A black body is an idealized object that absorbs all electromagnetic radiation (EMR) that falls on it.
- No EMR passes through it and none is reflected.
- Because no light (visible EMR) is reflected or transmitted, the object appears black when it is cold.
- This property is valid for radiation corresponding to all wavelengths and to all angles of incidence.
- Therefore, the black body is an ideal absorber of incident radiation.

Sun and Earth behave
approximately as black bodies



Conceptual Black Body

White Body

- All the incident radiation falling on the body are reflected.

Grey Body (Non Ideal Emitter/Imperfect Radiator)

- A body whose absorptivity of a surface does not vary with variation in temperature and wavelength of the incident radiation.
- A grey body will;
 - absorb some of the energy it will receive
 - reflect some of the energy it will receive
 - emit the energy it absorb

Black Body Radiation

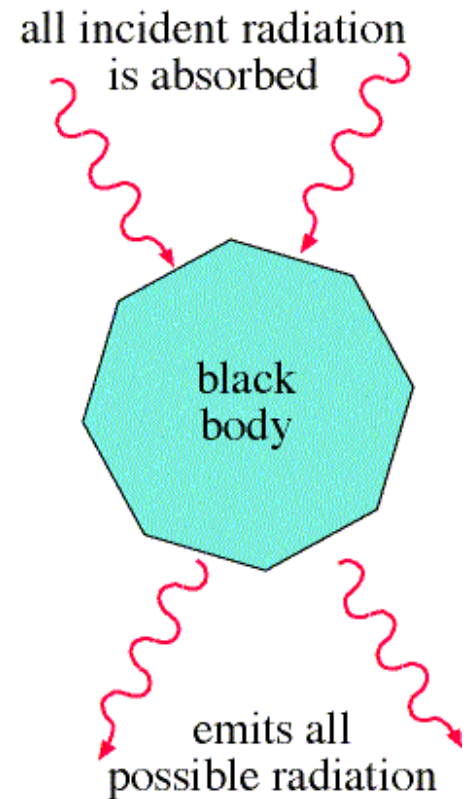
- A black body (BB) emits a temperature dependent spectrum of light.
- This thermal radiation from a black body is termed black-body radiation.
- At room temperature, BBs emit mostly infrared light.
- As the temperature increases BBs start to emit visible wavelengths, from red, through orange, yellow, and white before ending up at blue, beyond which the emission includes increasing amounts of UV.

Color Temperature of a Black-Body Radiator



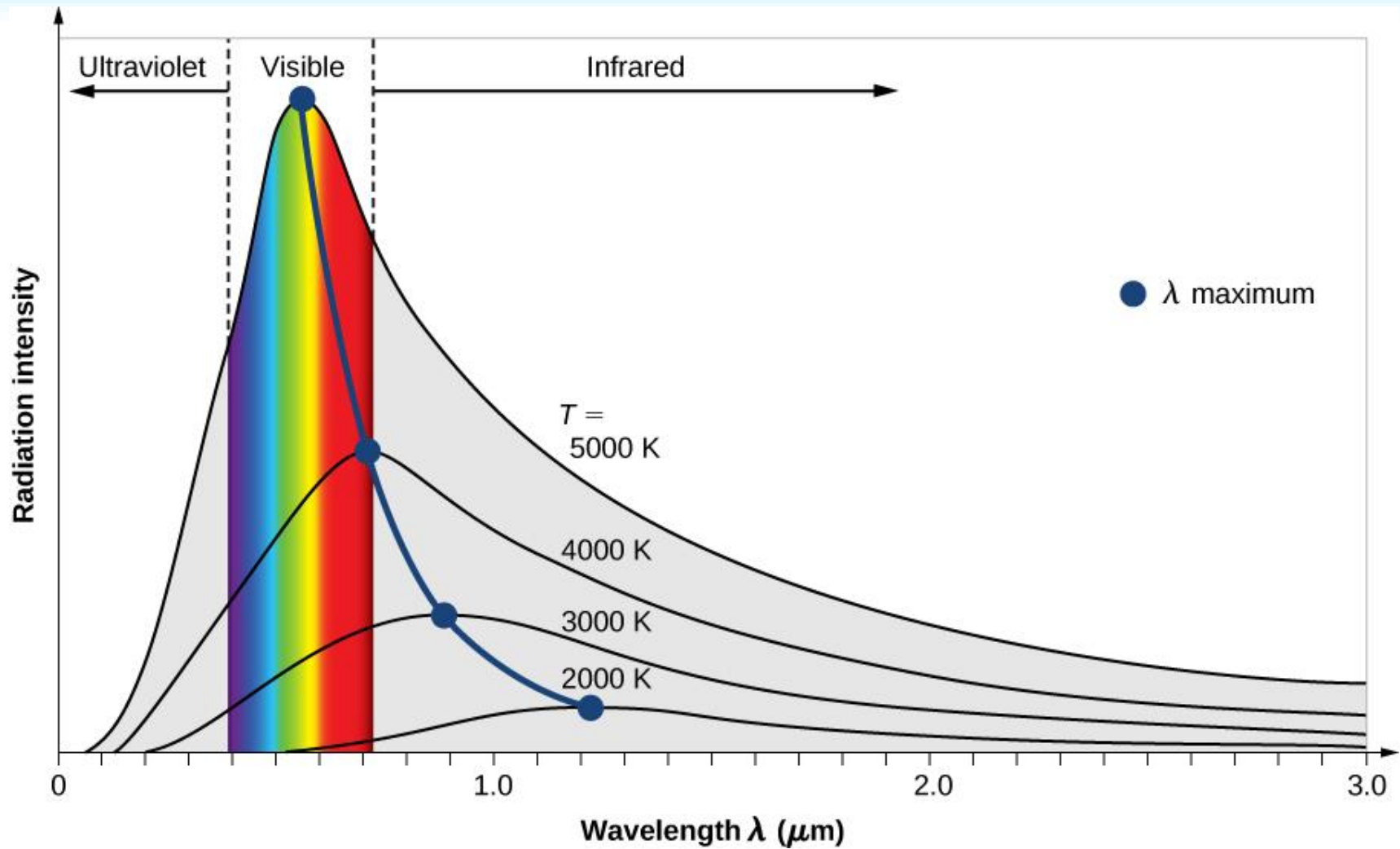
Black Body Radiation (cont.)

- At a particular temperature the black body would emit the maximum amount of energy possible for that temperature.
- Blackbody radiation does not depend on the type of object emitting it.
- Entire spectrum of blackbody radiation depends on only one parameter, the temperature, T .



Thermal radiation at equilibrium with an object at a fixed temperature

Black Body Radiation Spectra



Black Body Radiation Intensity (I)

- The intensity increases with increasing temperature.
- The amount of radiation emitted increases with increasing temperature.
 - The area under the curve
- The peak wavelength decreases with increasing temperature.
- The total energy radiated at T per unit time, per unit wave length and per unit area of the object.

Unit of I: $\text{Wm}^{-2}\text{nm}^{-1}$

Stefan-Boltzmann Law

- A black body radiates electromagnetic waves with a total energy flux F directly proportional to the fourth power of the Kelvin temperature T of the object.

$$F \propto T^4$$

$$\frac{P}{A} = F = \varepsilon \sigma T^4$$

$$P = \varepsilon \sigma A T^4$$

Where; F – Energy flux ($\text{Js}^{-1}\text{m}^{-2}$ / Wm^{-2})

σ - Stefan-Boltzmann constant ($5.67 \times 10^{-8} \text{ Wm}^{-2}\text{K}^{-4}$)

T – Object's temperature in Kelvin (K)

ε – Emissivity

A – Surface area (m^2)

P – Power radiated (W)

Stefan-Boltzmann Law (cont.)

Emissivity is defined as the ratio of the energy radiated from a material's surface to that radiated from a blackbody (a perfect emitter) at the same temperature and wavelength and under the same viewing conditions.

For an ideal black body: $\varepsilon = 1$

For an ideal white body: $\varepsilon = 0$

For a non ideal black body: $0 < \varepsilon < 1$

Stefan-Boltzmann Law (cont.)

Room temperature: T_0

Object's temperature: T

For an ideal black body ($\varepsilon = 1$);

The rate of emission of thermal radiation: $P_1 = \sigma A T^4$

The rate of absorption of thermal radiation: $P_2 = \sigma A T_0^4$

Resultant rate of emission of thermal radiation: $P = \sigma A (T^4 - T_0^4)$

For a non ideal black body ($0 < \varepsilon < 1$);

Resultant rate of emission of thermal radiation: $P = \varepsilon \sigma A (T^4 - T_0^4)$

Wien's Displacement Law

The wavelength at which the intensity peaks, λ_{max} , is inversely proportional to the absolute temperature.

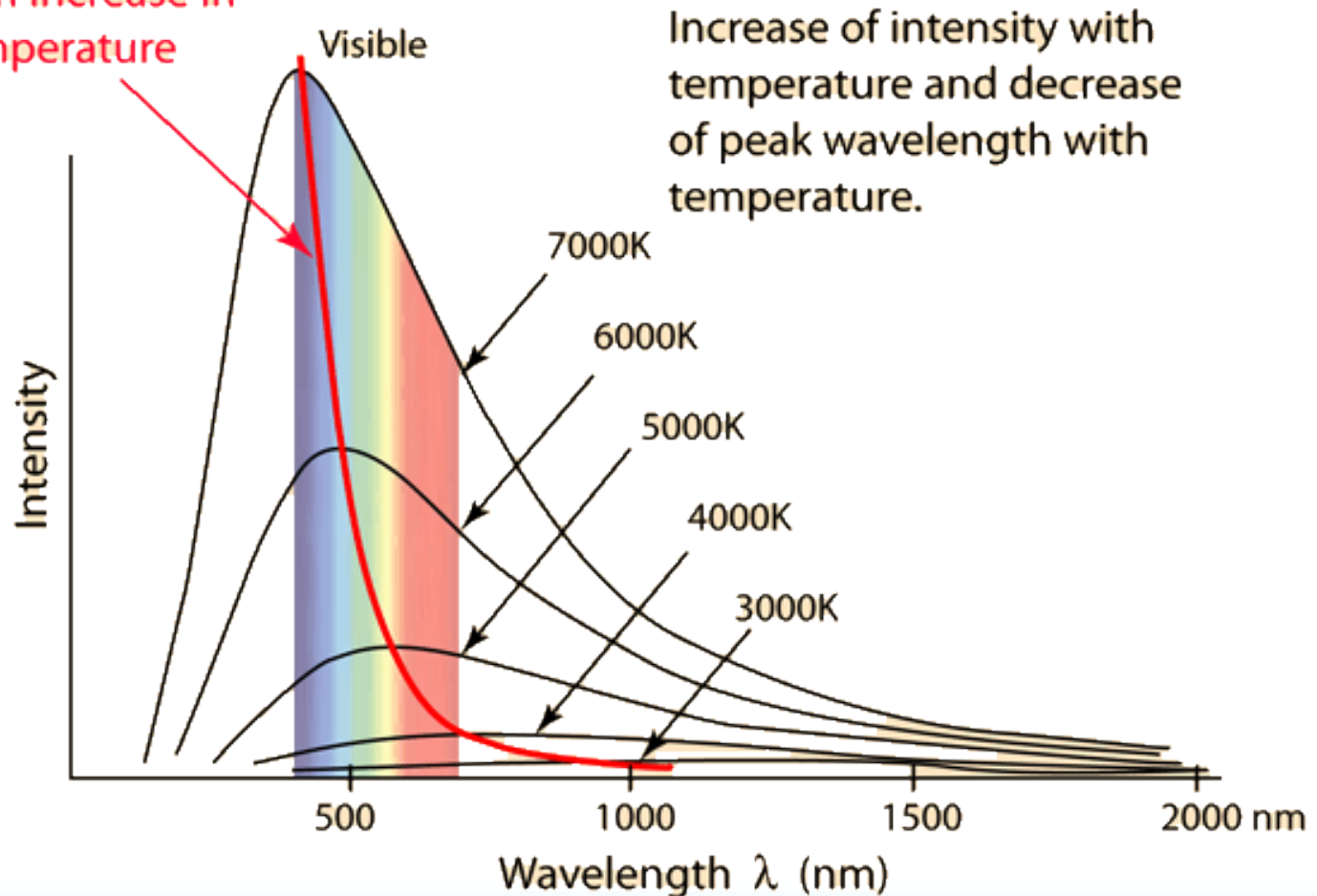
$$\lambda_{max} = \frac{b}{T}$$

$$\lambda_{m1} T_1 = \lambda_{m2} T_2$$

Where; λ_{max} - Peak wave length at absolute temperature T
b - Wien's displacement constant (2.898×10^{-3} mK)
T - Absolute temperature (K)

Wien's Displacement Law (cont.)

Decrease of λ_{peak}
with increase in
temperature



Applications of Wien's Displacement Law

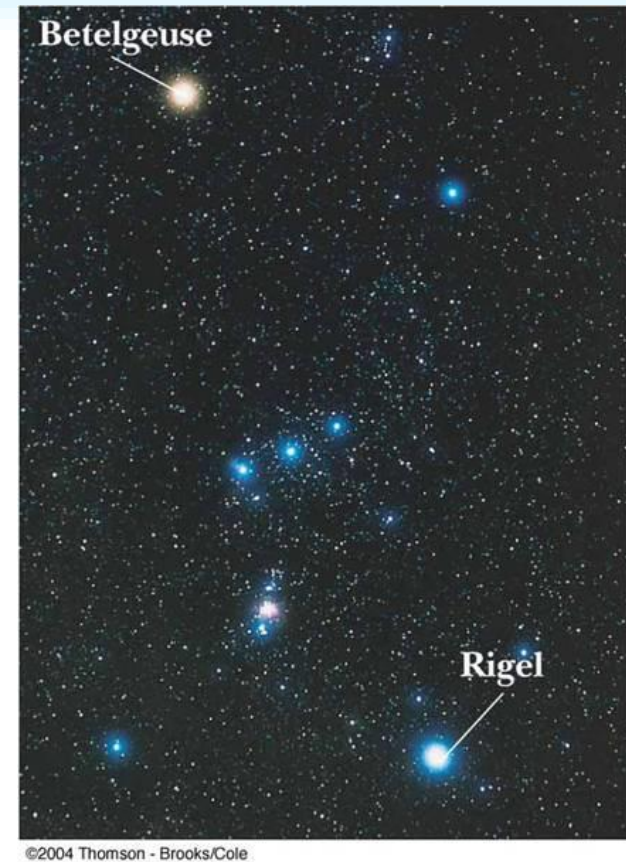
- Roughly we can say that the stars radiate like blackbody radiators.
- This is important because it means that we can use the theory for blackbody radiators to infer things about stars.

$$\lambda_{m1} T_1 = \lambda_{m2} T_2$$

- The wave length of a star can be measured by using a spectrometer.
- The temperature of the star surface can be determined by using the measured wave length and above equation.
- These secondary data can also be determined;
 - Chemical composition of the star
 - Age of the star

Applications of Wien's Displacement Law (cont.)

- This figure shows two stars in the constellation Orion. Betelgeuse appears to glow red, while Rigel looks blue in color. Which star has a higher surface temperature?



- The Sun approximates a 6000 K blackbody, its dominant wavelength (λ_{max}) is 0.48 μm .

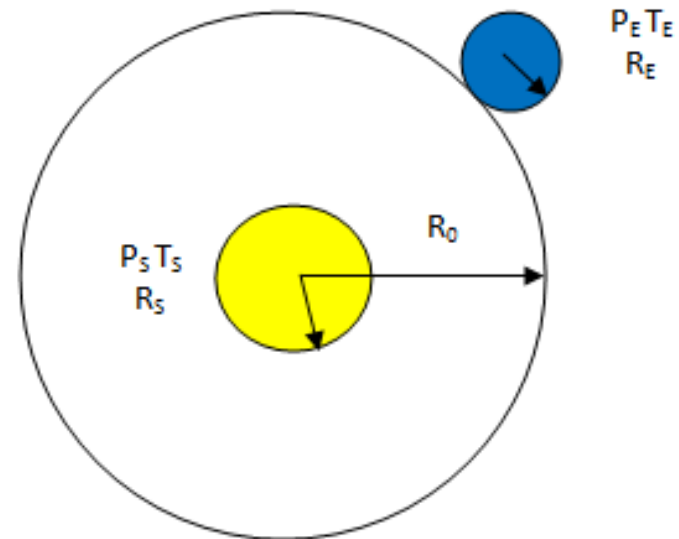
Thermal Equilibrium

If the rate of absorption of thermal radiation emitted by A from B is same for the emission of thermal radiation from B, then B is in thermal equilibrium with A.

At what temperature, the Earth is in thermal equilibrium with the Sun?

$$P = \varepsilon \sigma A T^4$$

For Sun and Earth $\varepsilon = 1$



Thermal Equilibrium (cont.)

Power radiated by Sun:

$$P_S = \sigma 4\pi R_S^2 T_S^4$$

Incident radiation on the Earth:

$$P_{E \text{ abso}} = \frac{P_S}{4\pi R_0^2} \times \pi R_E^2$$

Emission of thermal radiation from Earth:

$$P_{E \text{ emit}} = \sigma 4\pi R_E^2 T_E^4$$

If the Earth is in thermal equilibrium with the Sun: $P_{E \text{ abso}} = P_{E \text{ emit}}$

$$\frac{P_S}{4\pi R_0^2} \times \pi R_E^2 = \sigma 4\pi R_E^2 T_E^4$$

$$\frac{\sigma 4\pi R_S^2 T_S^4}{4\pi R_0^2} \times \pi R_E^2 = \sigma 4\pi R_E^2 T_E^4$$

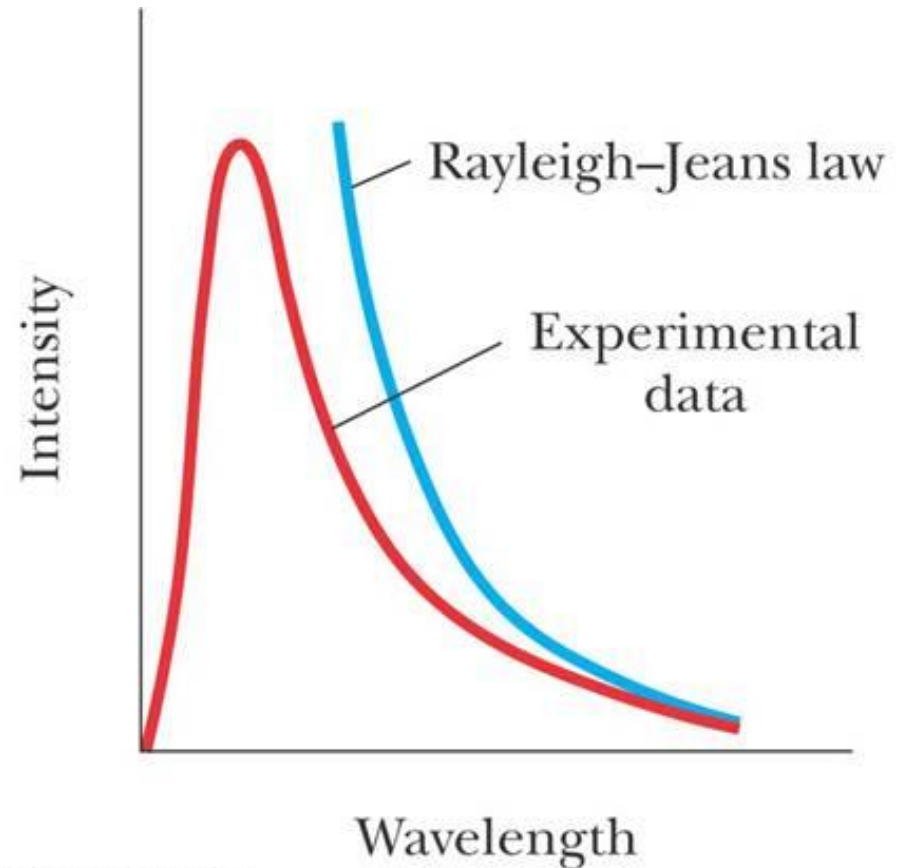
$$\frac{R_S^2 T_S^4}{4R_0^2} = T_E^4$$

Rayleigh-Jeans Law

- The Rayleigh–Jeans Law is an approximation to the spectral radiance of electromagnetic radiation as a function of wavelength from a black body at a given temperature through classical arguments.
- At long wavelengths, the law matched experimental results fairly well.

Rayleigh-Jeans Law (cont.)

- At short wavelengths, there was a major disagreement between the Rayleigh-Jeans law and experiment.
- This mismatch became known as the ultraviolet catastrophe.
 - You would have infinite energy as the wavelength approaches zero.



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Classical Physics

- Stefan-Boltzmann law, Wien's Displacement law, and Rayleigh-Jeans law are based on the concepts of classical physics.
- But Wien's Displacement law was exactly true for the short wave lengths and it was fairly disagree for long wave lengths.
- Rayleigh-Jeans law was true for long wave lengths and at short wave lengths, there was a major disagreement.

Planck's Theory of Blackbody Radiation

- In 1900 Planck developed a theory of blackbody radiation that leads to an equation for the intensity of the radiation.
- This equation is in complete agreement with experimental observations.
- He assumed the cavity radiation came from atomic oscillations in the cavity walls.
- Planck made two assumptions about the nature of the oscillators in the cavity walls.

Planck's Assumption, 1

- The energy of an oscillator can have only certain *discrete* values E_n

- $E_n = nhf$

$n=0,1,2,\dots$; n is called the quantum number

h is Planck's constant = 6.63×10^{-34} Js

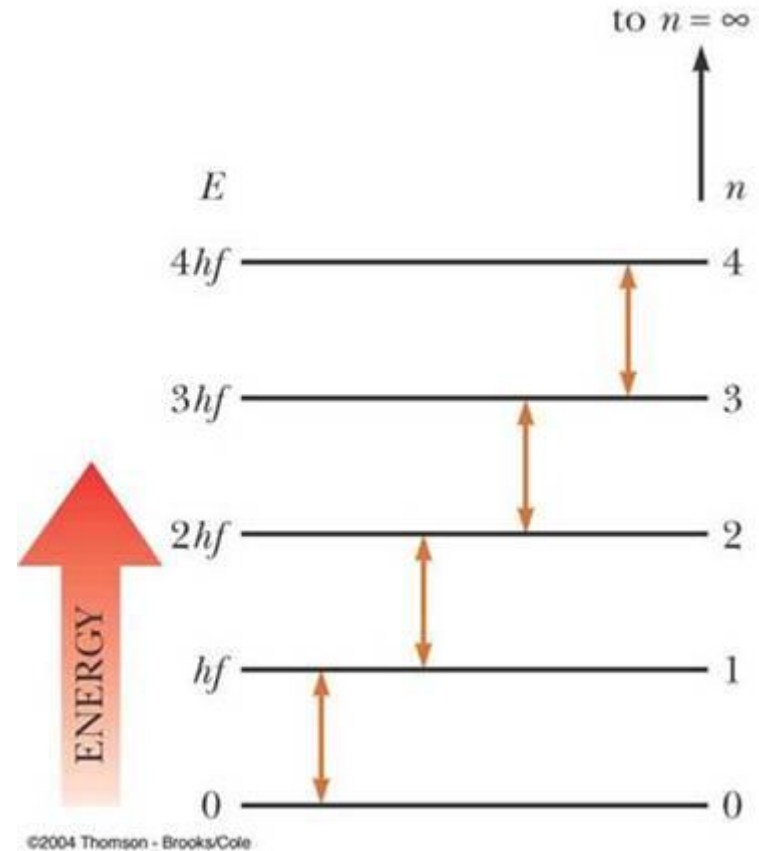
f is the frequency of oscillation

- The energy of the oscillator is **quantized**.
- Each discrete energy value corresponds to a different **quantum state**.

Each quantum state is represented by the quantum number, n

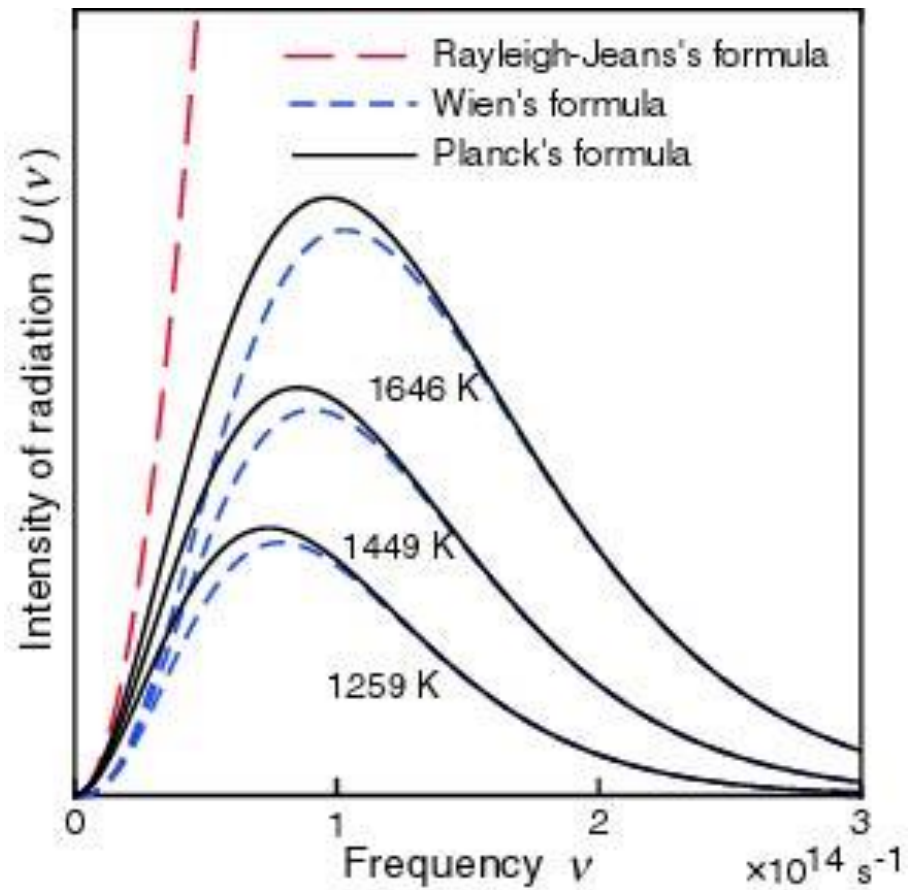
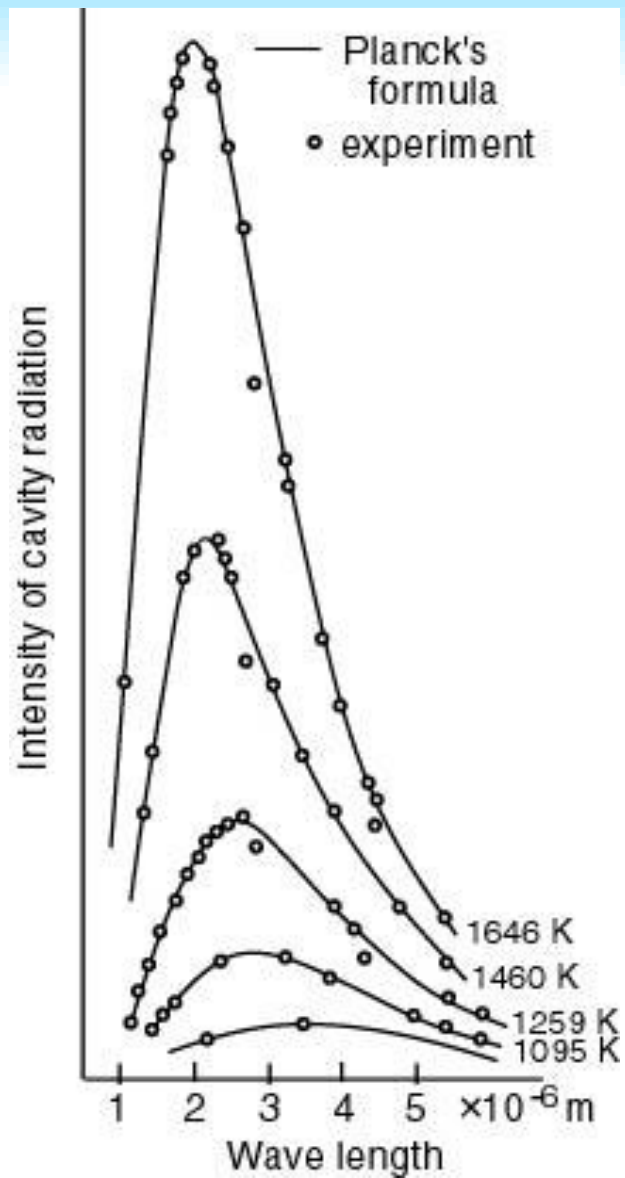
Energy Level Diagram

- An energy-level diagram of the oscillators showing the quantized energy levels and allowed transitions.
- Energy is on the vertical axis.
- Horizontal lines represent the allowed energy levels of the oscillators.
- The double-headed arrows indicate allowed transitions.



Planck's Assumption, 2

- The oscillators emit or absorb energy when making a transition from one quantum state to another.
- The entire energy difference between the initial and final states in the transition is emitted or absorbed as a single quantum of radiation.
- An oscillator emits or absorbs energy only when it changes quantum states.





Thanks for your attention

